Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

1 h 45 min · 90 marks · 7–9 questions · ≈ 1.2 min per mark. Every paper has a calculation, a graph and a 4–7 mark method.

TOP 12 MARK-LOSERS

- "Amount" — say **mass, volume or concentration**.
- Benedict's without **heating in a water bath**.
- Enzymes "killed" — say **denatured, active site changes shape**.
- Osmosis with "concentration of water" — use **water potential**.
- Method with no repeats, mean or control variables.
- Safety precaution with no reason.
- Graph: no units, small scale, missing key.
- Trend with no figures quoted.
- Units not converted (1 mm = 1000 µm); wrong d.p./s.f.
- Ligament ↔ tendon; left ↔ right chambers mixed up.
- "Fight infection", "scab", "diaphragm moves down".
- Genetic diagram missing gametes or phenotypes.

CELLS, TRANSPORT, ENZYMES

- Mitochondria: aerobic respiration, energy. Ribosomes: proteins. Bacteria: cell wall, plasmid, no nucleus.
- **Diffusion:** high → low concentration, passive. **Active transport:** low → high, ATP, carrier proteins. **Osmosis:** water, high → low water potential, partially permeable membrane.
- 3 cm cube: SA 54, V 27, SA:V 2 : 1.

ENZYMES AND TEMPERATURE

Low T: less kinetic energy → fewer collisions → fewer E–S complexes.

Above optimum: denatured → active site changes shape → substrate can't bind.

METHOD (6)

Substrate + enzyme in separate tubes in a water bath at set T → mix → gas syringe volume at set times → repeat at other T → same volume/concentration → repeat, mean.

FOOD AND DIGESTION

Test	Positive
Benedict's + heat	blue → brick red
iodine	brown → blue-black
biuret	blue → lilac
ethanol, then water	cloudy white

- Energy (J) = mass water × 4.2 × ΔT: 28 × 12 × 4.2 = 1411 J. Errors: heat loss, incomplete burning.
- Protein 17 kJ/g, lipid 37 kJ/g: 1.2 × 17 + 4.8 × 37 = 198 kJ.
- Vit C scurvy · iron haemoglobin · vit D calcium uptake · vit A eye pigment.
- Bile (liver → gall bladder): emulsifies fat → larger SA for lipase; neutralises acid.
- Peristalsis: circular muscles contract behind bolus; fibre adds bulk.
- BMI = mass ÷ height^2: 67.0 ÷ 1.69^2 = 23.5 (healthy).

BLOOD AND CIRCULATION

Part	Job
red cell	haemoglobin carries O_2
phagocyte	engulfs pathogens
lymphocyte	antibodies
platelet	clotting
plasma	dissolved substances, heat

CLOTTING (5)

Damaged platelets → thromboplastin → prothrombin → thrombin → fibrinogen → fibrin mesh → traps red cells → clot (Ca^{2+}). Stops blood loss and pathogen entry.

- Left ventricle thickest: pumps to whole body at high pressure. Valves stop backflow.
- Pressure: highest in aorta; arteries recoil (elastic); falls in capillaries → smooth flow for exchange.
- Tissue fluid: arterial end 4 − 3.3 = +0.7 kPa out; venous 1.6 − 3.3 = −1.7 kPa in; proteins stay in.
- CHD: atheroma in coronary arteries → less O_2/glucose to heart muscle.
- Exercise: more blood to muscles and skin, less to gut.

BREATHING

INHALATION (6)

Intercostals contract → ribs up and out; diaphragm contracts and **flattens** → thorax volume ↑ → pressure ↓ → air drawn in.

- Alveoli: large SA, thin wall, capillaries, moist. Emphysema: walls break → less SA.
- Tidal (normal) · vital capacity (max in → max out) · residual (left in lungs).
- More CO_2 → chemoreceptors → faster, deeper breathing.
- $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ energy).

KIDNEY AND LIVER

- Ultrafiltration: high pressure in glomerulus (wide in, narrow out) → small molecules into capsule; cells and proteins stay.
- Glucose reabsorbed by active transport; water by osmosis (ADH).

HOT DAY (5)
Sweat ↑ → blood water potential ↓ → hypothalamus → pituitary releases ADH → collecting duct more permeable → more water reabsorbed → small volume, concentrated urine.
Alcohol ↓ ADH → large volume, dilute urine.

- Dialysis: fluid has no urea → urea diffuses out across partially permeable membrane; heparin stops clots.
- Liver: deamination → urea; detoxifies alcohol, H_2O_2; bile; glycogen store.

NERVES AND SENSES

SYNAPSE (6)
Impulse reaches axon end → vesicles release neurotransmitter → diffuses across cleft → binds receptors on post-synaptic membrane → new impulse.

- Sensory: receptor → CNS, cell body in middle. Motor: CNS → effector, long axon, cell body at end.
- Reflex: rapid, automatic, protective. Myelin insulates and speeds impulses.
- Speed = distance ÷ time: 0.25 ÷ 0.0045 = 55.6 m/s.
- Near object: ciliary contracts → ligaments slacken → lens thicker → more refraction.
- Short sight: concave lens. Long sight: convex. Cataract: lens replaced.
- Ossicles amplify; cochlea hair cells; Eustachian tube equalises pressure; semicircular canals = balance.

HOMEOSTASIS

DEFINITION (3)
Constant internal environment despite external changes, by negative feedback.

- Hot: hypothalamus → vasodilation → radiation; sweat evaporates.
- Cold: shivering, vasoconstriction, less sweat, hairs erect.
- Too hot: enzymes denatured → reactions stop.
- High glucose → insulin (pancreas) → glycogen in liver/muscle. Low → glucagon.
- Adrenaline: heart rate ↑, airways dilate, glycogen → glucose.

SKELETON AND MUSCLE

- Cartilage (friction, shock) · synovial fluid (lubricates) · ligament (bone–bone) · tendon (muscle–bone).
- Elbow/knee hinge; hip/shoulder ball and socket.
- Arm up: biceps contracts, triceps relaxes (antagonistic).
- Cardiac: striated, branched. Skeletal: striated, many nuclei. Smooth: unstriated, spindle.

REPRODUCTION

- Oestrogen repairs the lining; ovulation monthly; progesterone maintains lining.
- Sperm: tail, many mitochondria (energy to swim), acrosome enzymes, haploid nucleus.
- Meiosis: haploid (23) → diploid (46) restored → variation.
- Vasectomy: vas deferens cut → no sperm → no fertilisation; prostate fluid can still carry HIV.
- XX × XY → 50% each sex.

DNA AND INHERITANCE

DNA (4)
Double helix · nucleotides (phosphate, deoxyribose, base) · A–T, C–G · hydrogen bonds.
RNA: single strand, ribose, uracil.

- Mitosis: 2 identical diploid cells (growth). Meiosis: 4 different haploid gametes.
- Mutation: base sequence → amino acids → protein shape. UV causes skin cancer.
- I^A, I^B codominant; I^o recessive. Group O: no antigens. Mismatch → agglutination.
- Sex-linked: $X^DX^d \times X^DY$ → 1/4 affected. Recessive: skips generations.
- Insulin GM: restriction enzymes cut gene and plasmid → ligase → recombinant plasmid → bacteria → fermenter.

DISEASE AND IMMUNITY

VACCINATION (5)
Weakened pathogen/antigen → lymphocytes make antibodies → memory cells → fast response on reinfection.

RESISTANCE (5)
Mutation → resistant bacteria survive → reproduce → pass on; overuse/unfinished courses.

- Antibiotics don't work on viruses (HIV). HIV = virus; AIDS = disease.
- Growth curve: lag → exponential → stationary → death.
- Aseptic: wash hands, flame loop, lid taped, ≤ 25 °C.
- Clear zone diameter: larger = more effective antibiotic.

CALCULATIONS

Type	Example
magnification	64 000 µm ÷ 2.0 µm = ×32 000
actual size	20 mm ÷ 750 = 26.7 µm
% change	(2629 − 2103) ÷ 2103 × 100 = 25%
2 s.f.	212.5% → 210%
ratio 1 : n	7500 : 13 750 = 1 : 1.83
rate	0.5 ÷ 96 = 0.0052 AU/min
standard form	70 000 000 ÷ 5000 = 1.4×10^4